

Class #5 - Complex eigenvalues

1. Example: Let the matrix $A = \begin{bmatrix} 1 & -2 \\ 1 & 3 \end{bmatrix}$ acts on $\mathbf{C}^2$. Find the eigenvalues and a basis for each eigenspace in $\mathbf{C}^2$.

2. The transformation $x \mapsto Ax$ is the composition of a rotation and a scaling. Give the angle θ of the rotation, where $-\pi < \theta \leq \pi$, and give the scale factor γ .

$$A = \begin{bmatrix} \sqrt{3} & 3 \\ -3 & \sqrt{3} \end{bmatrix}$$

3. Note: If $A = \begin{bmatrix} a & -b \\ b & a \end{bmatrix}$, where a and b are _____ and not both zero, then the eigenvalues are _____. If we let $\gamma =$ _____, then

$$A =$$

The angle θ is called the *argument* of $\lambda = a + bi$.

4. It is true that for any real 2×2 matrix with a complex eigenvalue, it is actually a composition of rotation and scaling after an appropriate change of variables.
5. Theorem: Let A be a real 2×2 matrix with a complex eigenvalue $\lambda = a - bi$ ($b \neq 0$) and an associated eigenvector v in $\mathbf{C}^2$. Then,

$$A = PCP^{-1}, \quad \text{where } P = [\operatorname{Re} v \quad \operatorname{Im} v] \quad \text{and} \quad C = \begin{bmatrix} a & -b \\ b & a \end{bmatrix}.$$

Proof:

- (1) For any vector x in $\mathbf{C}^2$, $\operatorname{Re}(Ax) = A(\operatorname{Re} x)$, and $\operatorname{Im}(Ax) = A(\operatorname{Im} x)$.

$$(2) \ A(\operatorname{Re} v) = a\operatorname{Re} v + b\operatorname{Im} v \text{ and } A(\operatorname{Im} v) = -b\operatorname{Re} v + a\operatorname{Im} v$$

$$(3) \text{ Conclude } AP = PC.$$

$$6. \text{ Example: } A = \begin{bmatrix} 1 & -1 & 0 \\ 1 & 1 & 0 \\ 0 & 0 & 2 \end{bmatrix}$$